

Fibonacci: Getting Rich Using Math

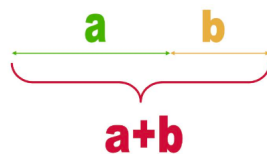
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1 Introduction

Leonardo Bonacci, better known as Fibonacci, was an Italian-born mathematician. He is oftentimes regarded as one of the most influential and talented Western mathematicians of his time. Fibonacci traveled much of his youth, constantly interacting with merchants and traders. He met merchants from all along the Mediterranean coast, each with their own unique system of performing arithmetic. It was in Algeria where he discovered the Hindu-Arabic numeral system. In comparison to the Roman numerals he was used to, I II III IV, and so on, the Hindu-Arabic system was far more efficient for computations. The Hindu-Arabic numeral system is one that we use to this day, with 10 digits, 0 through 9. Fibonacci's love for numbers, combined with knowledge learned from the merchants he met in his travels inspired him to write *Liber Abaci*. *Liber Abaci* is widely recognized for its broadcasting of Hindu-Arabic numerals to western numerals. In the mathematical book, he discusses computations with this new numerical system, and proposes solutions to mathematical puzzles. This paper will discuss Fibonacci's influence on the Western world through his work in *Liber Abaci*, the connection between the Fibonacci sequence and the Golden Ratio, as well as real-world applications of the Fibonacci sequence.

2 Defining the Golden Ratio

If two quantities are in a Golden Ratio then their ratio is equal to the ratio of the larger of them to the sum. Let's take a look at the image below:



if segments a and b were to be in a Golden Ratio, $\phi = \frac{a}{b} = \frac{a+b}{a}$ (1). Let's see if we can represent (1) in a different form: $\frac{a}{b} = \frac{a}{a} + \frac{b}{a}$ (2). On the left side of (2) we have the ratio of $\frac{a}{b}$, which we have denoted ϕ . On the right side we have $\frac{b}{a}$, which is equal to the inverse of $\frac{a}{b}$, $\frac{a}{b}^{-1} = \phi^{-1} = \frac{1}{\phi}$. Lets plug this new found information into (2): $\phi = 1 + \frac{1}{\phi}$ (3). Now, let's try and solve (3). We can transform (3) into a quadratic equation. Multiply (3) by ϕ to get rid of it in the denominator

under the condition that $\phi \neq 0$,

$$\begin{aligned}\phi &= 1 + \frac{1}{\phi} \\ \phi^2 &= \phi + 1 \\ \phi^2 - \phi - 1 &= 0\end{aligned}$$

Let's call $\phi^2 - \phi - 1 = 0$ (4). We can now use the quadratic equation to solve (4). Using the quadratic formula,

$$\begin{aligned}\phi_{1,2} &= \frac{1 \pm \sqrt{(-1)^2 - 4(1)(-1)}}{2(1)} \\ \phi_{1,2} &= \frac{1 \pm \sqrt{1+4}}{2}\end{aligned}$$

where $\phi_1 = \frac{1+\sqrt{5}}{2}$ (6) and $\phi_2 = \frac{1-\sqrt{5}}{2}$ (7). ϕ_1 is noted as the Golden Ratio and is ≈ 1.618 . Evidently ϕ_2 is less than zero.

The Golden Ratio can also be found through a continued fraction. Let's consider the following continued fraction,

$$d = 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1 + \dots}}}$$

Let's label this (8) and take a look at the first few convergents of d:

$$\begin{aligned}d_0 &= 1 \\ d_2 &= 1 + \frac{1}{1+1} = 1.5 \\ d_3 &= 1 + \frac{1}{1 + \frac{1}{1+1}} = 1 + \frac{1}{d_2} = 1.66667 \\ d_4 &= 1 + \frac{1}{1 + \frac{1}{1 + \frac{1}{1+1}}} = 1 + \frac{1}{d_3} = 1.625\end{aligned}$$

Notice the denominator of the next term is the value of the previous term. Here are the values of the partial sums up until d_{14} :

Proof of Continued Convergence	
d_0	1
d_1	1.5
d_2	1.666666667
d_3	1.6
d_4	1.625
d_5	1.615384615
d_6	1.619047619
d_7	1.617647059
d_8	1.618181818
d_9	1.617977528
d_{10}	1.618055556
d_{11}	1.618025751
d_{12}	1.618037135
d_{13}	1.618032787
d_{14}	1.618034448

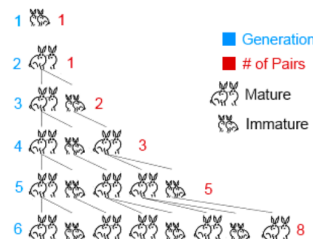
Around d_{10} is when you can see 1.618 emerge, or as we have previously discovered, the Golden Ratio, ϕ . The further we calculate and move toward d_∞ the evermore close we get to 1.618. In (7) notice since the fraction is infinite the lower part of the fraction is equivalent to the upper. Because of this, we can rewrite (7) as $d = 1 + \frac{1}{d}$. Which we can transform into a quadratic equation of the form $d^2 = d + 1$ or $d^2 - d - 1 = 0$ (9). By comparing (4) and (9), we have derived the quadratic equation which leads to the Golden Ratio in two different ways.

3 The Rabbit Island Problem

Your second section goes here. The real question is, how does any of this have relevance in our everyday life. The rabbit population may not be something that often crosses your mind, but this was something it was widely thought about by merchants in the 12th and 13th centuries. This brought about the problem of Rabbit Island. On pages 283 through 284 of *Liber Abaci*, Fibonacci ponders: In theory, if one started with a pair of rabbits, male and female, how many rabbits would they produce after one year? There were a few rules based on ideal circumstances to this puzzle:

1. No rabbit can die, or be consumed by a predator.
2. Each female rabbit must reproduce every month after her second month being alive.
3. Every time the rabbits reproduce, the female gives birth to a pair, one male and one female rabbit.

To help better visualize this problem, take a look at the diagram below:



For reference $f(n)$ represents the amount of pairs of rabbits at the beginning of any month, n . Additionally the function $g(n)$ marks the total number of rabbit pairs after n generations. One of the rules we must follow is that the rabbits must be mature, or above 2 months to mate. This means that in generation n , a rabbit born $n - 2$ generations ago can produce offspring. We can model this as rabbits producing $= f(n - 2)$. From this, the first time a new pair of rabbits will be produced is generation 3. Assuming the rabbits live indefinitely, there will always be $g(n - 1)$ rabbits at any given time prior to a new generation.

We can combine the two expressions of finding independent aspects of the rabbit population into one equation that notates the entirety of the rabbit population.

In plain English, the total rabbit population is equal to the sum of the amount of rabbits able to produce with general rabbits. From this statement we can construct the general equation:

$$f(n) = f(n - 1) + f(n - 2)$$

Where $n > 2$ since the rabbits can only reproduce after month 2.

By now, you might recognize this equation as well as the sequence of rabbit pairs. The problem we just observed, the Rabbit Island, is what birthed the Fibonacci Sequence.

The Fibonacci sequence, to me, has always seemed fairly simple. If anything, it is something younger me thought I had created. Each term of the sequence can be noted as a sum of the previous two.

The sequence starts with the number $f(1) = 1$, the next term would be $f(2) = 1 + 0 = 1$, the next being $f(3) = 1 + 1 = 2$, the following term would be $f(4) = f(3) + f(2) = 2 + 1 = 3$. Do these numbers look familiar, hence the rabbit problem. We can generalize this as $f_n = f_{n-1} + f_{n-2}$ for $n > 1$.

Let's take a closer look at the ratio between two consecutive numbers in the Fibonacci sequence:

Term #	Fibonacci #	Ratio $\frac{f_{n+1}}{f_n}$
1	0	N/A
2	1	N/A
3	1	1
4	2	2
5	3	1.5
6	5	1.666666667
7	8	1.6
8	13	1.625
9	21	1.615384615
10	34	1.619047619
11	55	1.617647059
12	89	1.618181818
13	144	1.617977528

14	233	1.618055556
15	377	1.618025751
16	610	1.618037135
17	987	1.618032787
18	1597	1.618034448
19	2584	1.618033813
20	4181	1.618034056
21	6765	1.618033963
22	10946	1.618033999
23	17711	1.618033985
24	28657	1.61803399

It is evident that as the sequence progresses, the ratio approaches ϕ , the Golden Ratio. From this we can see that the ratio of two consecutive Fibonacci numbers is equal to the golden ratio:

$$\lim_{n \rightarrow \infty} \frac{f(n+1)}{f(n)} = \phi$$

4 Proof Through Induction

Let's see if we can generalize Fibonacci's n th term, $f(n)$, using the Golden Ratio. To better understand this, let's look at (4): $\phi^2 - \phi - 1 = 0$. We can rewrite this equation as $\phi^2 = \phi + 1$ (10). From this we can find expressions for all powers of ϕ by mathematical induction. Mathematical induction works through proving the base case, $n=1$, and then applying that to $n = k$, and proving it must hold for all future cases, $n = k + 1$:

$$\begin{aligned} \phi &= \phi \\ \phi^2 &= \phi + 1 \\ \phi^3 &= \phi(\phi^2) \rightarrow \phi^3 = \phi(\phi + 1) \rightarrow \phi^3 = \phi^2 + \phi \rightarrow \phi^3 = (\phi + 1) + \phi \rightarrow \phi^3 = 2\phi + 1 \\ \phi^4 &= \phi^3 \times \phi \rightarrow \phi^4 = (2\phi + 1)\phi \rightarrow \phi^4 = 2\phi^2 + \phi \rightarrow \phi^4 = 2(\phi + 1) + \phi \rightarrow \phi^4 = 3\phi + 2 \\ \phi^5 &= \phi^4 \times \phi \rightarrow \phi^5 = (3\phi + 2)\phi \rightarrow \phi^5 = 3\phi^2 + 2\phi \rightarrow \phi^5 = 3(\phi + 1) + 2\phi \rightarrow \phi^5 = 5\phi + 3 \\ \phi^6 &= \phi^5 \times \phi \rightarrow \phi^6 = (5\phi + 3)\phi \rightarrow \phi^6 = 5\phi^2 + 3\phi \rightarrow \phi^6 = 5(\phi + 1) + 3\phi \rightarrow \phi^6 = 8\phi + 5 \end{aligned}$$

And so forth. Hence we get $n = f_n + f_{n-1}$ (11), where f_n and f_{n-1} are consecutive numbers of the Fibonacci sequence.

We know from equations $\phi_1 = \frac{1+\sqrt{5}}{2}$ and $\phi_2 = \frac{1-\sqrt{5}}{2}$, ϕ_1 and ϕ_2 are the roots of the equation $\phi^2 = \phi + 1$. That means that they must satisfy $\phi^n = f_n\phi + f_{n-1}$ (11). This is because it was derived inductively. Ergo, we can substitute each root into equation (11) one at a time. Which gives us the following: $\phi_1^n = f_n\phi_1 + f_{n-1}$ for ϕ_1 (12), and $\phi_2^n = f_n\phi_2 + f_{n-1}$ for ϕ_2 (13). Let's subtract (13) from (12):

$$\phi_1^n - \phi_2^n = (f_n\phi_1 + f_{n-1}) - (f_n\phi_2 + f_{n-1}) \quad (14)$$

Let's simplify (14):

$$\begin{aligned}\phi_1^n - \phi_2^n &= f_n(\phi_1 - \phi_2) + f_{n-1} - f_{n-1} \\ \phi_1^n - \phi_2^n &= f_n(\phi_1 - \phi_2) \quad (15)\end{aligned}$$

Let's solve (15) for f_n :

$$f_n = \frac{\phi_1^n - \phi_2^n}{\phi_1 - \phi_2}$$

Now using the actual values discovered in (6) and (7), we get the final formula for the n th term of Fibonacci sequence in closed form:

$$f_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}$$

For example, let's say we wanted to find the 12th term of the Fibonacci sequence. We could use the formula above to do so:

$$\begin{aligned}f_{12} &= \frac{\left(\frac{1+\sqrt{5}}{2}\right)^{12} - \left(\frac{1-\sqrt{5}}{2}\right)^{12}}{\sqrt{5}} \\ f_{12} &= 144\end{aligned}$$

This is in fact true.

5 Applications in Finance

This core idea seems elementary, so why is it so important? Fibonacci's understanding of rabbit populations, the golden ratio, and other discoveries in *Liber Abaci*, can be applied to a variety of things, both real world and mathematical. But this is not the only interesting observation that can be made about the Fibonacci sequence. When I was younger, I would sometimes come with my dad to work. I would stare at the huge computer screens, filled with strange charts and numbers. The world of finance has always been something I was interested in, but have never quite understood. Fibonacci's ratio can be used in finance to help traders identify when to stop losses, and target prices for an amicable price.

As previously shown,

$$\lim_{n \rightarrow \infty} \frac{f(n+1)}{f(n)} = 1.618 = \phi$$

But this is not the only ratio that converges to a constant. For example,

$$\lim_{n \rightarrow \infty} \frac{f(n+2)}{f(n)} = 2.618$$

$$\lim_{n \rightarrow \infty} \frac{f(n+3)}{f(n)} = 4.236$$

$$\lim_{n \rightarrow \infty} \frac{f(n+4)}{f(n)} = 6.854$$

The inverses of these first four limits of the Fibonacci sequence are called retracement levels in finance. People who work with this analysis are called technical analysts. Let's work out an example to better understand how they are actually used.

To find the retracement levels we must take the inverses of the four limits as follows:

Retracemnts Levels for Stocks	
ratio $\frac{f_n}{f_{n+1}}$	0.6181818182
ratio $\frac{f_n}{f_{n+2}}$	0.3823529412
ratio $\frac{f_n}{f_{n+3}}$	0.2363636364
ratio $\frac{f_n}{f_{n+4}}$	0.1459016393

For the purpose of this paper, let's focus on IBM stock. I have gone ahead and created a graph of the stock, as well as marked the retracement levels accordingly, and circled any main focus points that I will discuss later.

To help understand the graph:

$r_0 = 113.26$, which is 0% retracement level or the lowest value of the stock price.

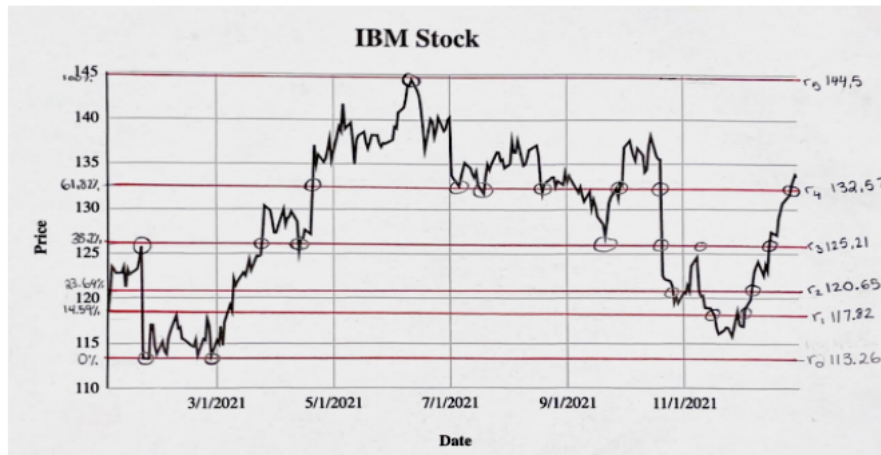
$r_5 = 144.5$, which is 100% retracement level or the highest value of the stock price.

$r_1 = r_0 + (r_5 - r_0) \times \frac{f_n}{f_{n+4}}$. By plugging in the actual numbers you get:

$$r_1 = 113.26 + (144.5 - 113.26) \times 0.1459016393 = 117.82$$

We can derive r_2 , r_3 , and r_4 this way. And they respectively are equal to:

$$r_2 = 120.65, r_3 = 125.21, r_4 = 132.57.$$



Let's consider the period of time between January 1, 2021 and May 1, 2021. As we can see the stock price in January tries to grow higher in value from r_1 , passing r_2 , but finding resistance at level at r_3 . Then between around February and March, it trades in the range of r_0 and r_1 . What I find cool about this, is that it perfectly touches r_1 but finds resistance and never passes it. From around April to May, the stock price breaks r_3 and floats between r_3 and r_4 before eventually breaking r_4 . After May 1, 2021, I notice the stock price precisely hit r_4 three times before breaking down. This would be a good time to sell. The main takeaway is: if it breaks up, buy. If it breaks down, sell.

Although the Fibonacci sequence was developed in the 13th century it is still extremely applicable and relevant today- this is what makes it so beautiful. Whether or not Fibonacci retracement

works in relation to stock prices, or it is simply a psychological phenomenon, remains a curiosity of mine. The applications of Fibonacci are so much more complex than I could have ever imagined. And hey, maybe now I can predict some stock and become rich.

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